Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Hardware Trade Study

Reference	TBD **[TBD-1]**
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1.0 Purpose and Scope

This study chooses the hardware of the bench on which the System is built and verified. It measures each candidate against the hardware needs of the architecture [2, Sec. 8.0] and the bench needs of the verification plan [3, Sec. 5.0], and prefers hardware the author already owns. It chooses in two steps: a first bench of owned boards only, and the parts the full bench adds.

2.0 Documents

[1] NASA, *NASA Systems Engineering Handbook*, NASA/SP-2016-6105 Rev 2, 2016.
[2] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Architecture*, Issue 1, Revision 0, Draft.
[3] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Verification Plan*, Issue 1, Revision 0, Draft.
[4] STMicroelectronics, *STM32G431xB Datasheet*, DS12589 Rev 6.
[5] STMicroelectronics, *STM32G4 Series Reference Manual*, RM0440 Rev 9.
[6] STMicroelectronics, *STM32G4 Nucleo-32 Board (MB1430)*, UM2397 Rev 2.
[7] STMicroelectronics, *Introduction to System Memory Boot Mode on STM32 MCUs*, AN2606 Rev 70.
[8] STMicroelectronics, *Discovery Kit with STM32MP257F MPU*, UM3385 Rev 2.
[9] Texas Instruments, *THVD14xx 3.3-V to 5-V RS-485 Transceivers*, SLLSEY3E, 2019.
[10] Texas Instruments, *TCAN33x 3.3V CAN Transceivers with CAN FD*, SLLSEQ7F, 2025.
[11] Texas Instruments, *TMUX1574 2:1 (SPDT) 4-Channel Switch*, SCDS391C, 2019.
[12] Linux kernel, `drivers/net/can/usb/gs_usb.c`, mainline, read 12 October 2026.
[13] `uhubctl` project, *README*, read 12 October 2026.
[14] STMicroelectronics, *STM32G474xB/xC/xE Datasheet*, DS12288 Rev 3 (distributor copy; later revisions not obtained).
[15] STMicroelectronics, *STM32G491xC/xE Datasheet*, DS13122 Rev 4 (distributor copy).
[16] STMicroelectronics, *STM32G4 Nucleo-64 Boards (MB1367)*, UM2505 Rev 4 (distributor copy; Rev 7 not obtained).
[17] D. Owens et al., "Efficient and Side-Channel Resistant Ed25519 on ARM Cortex-M4", *IEEE Trans. Circuits and Systems I*, vol. 71, no. 6, June 2024.

3.0 Criteria

Each need of the architecture and of the verification plan is a criterion. Each candidate gets one of four answers for each criterion. Among candidates that meet the needs, owned hardware comes first, then the lowest cost.

Legend. MET: met as the board is. PARTLY: met only with a board change, or with a limit stated beside it. NOT MET: not met. TO MEASURE: settled only by a measurement on the bench [3, Sec. 7.6].

4.0 Controller

Three boards are assessed: the NUCLEO-G431KB, which the author owns, and two Nucleo-64 boards built on the same board design, the NUCLEO-G474RE and the NUCLEO-G491RE [16, p. 1]. A published Ed25519 verification on a Cortex-M4 takes 720 000 cycles [17, p. 2674]; two take about 8.5 ms at 170 MHz, a figure measured on the bench for each board [3, Sec. 7.6].

Need [2, Sec. 8.0]	NUCLEO-G431KB	NUCLEO-G474RE	NUCLEO-G491RE
Clock and RAM for two signature checks	TO MEASURE: 170 MHz, 32 KB SRAM [4, pp. 1, 14]	TO MEASURE: 170 MHz, 128 KB SRAM [14, Sec. 3.5, p. 20]	TO MEASURE: 170 MHz, 112 KB SRAM [15, Sec. 3.5, pp. 18–19]
Flash; writing while running	PARTLY: 128 KB, one bank; a page erase stalls code for up to 24.47 ms [5, p. 195; 4, p. 107]	MET: 512 KB in two banks, read-while-write with the DBANK option bit set [14, Sec. 3.4, p. 19; 5, Sec. 3.3.8, p. 108]	PARTLY: 512 KB, one bank; erase stalls code [5, p. 156]
UART for data link A	PARTLY: USART2 wired to the ST-LINK by default [6, p. 16]	MET: UART5, apart from the factory loader's UARTs [14, p. 75]	MET: UART5 [15, Table 13, p. 65]
CAN FD controller	MET: one [4, Table 13, p. 61]	MET: three [14, Table 2, p. 15]	MET: two [15, p. 14]
SWD with full memory access	PARTLY: reaches only the on-board ST-LINK [6, pp. 22–23]	MET: PA13/PA14 on header CN7 pins 13 and 15 [16, Table 16, p. 31]	MET: same board as the G474RE [16, Table 16, p. 31]
Factory loader on its own UART	MET: USART1 [7, Table 115, p. 274]	MET: USART1 to 3 [7, Table 117, p. 279]	MET: USART1 to 3 [7, Table 119, p. 285]
NRST and BOOT pins accessible	PARTLY: BOOT0 not on a header [6, p. 16]	PARTLY: NRST on CN7 pin 14; BOOT0 on CN7 pin 7 but not connected by default [16, Table 16, pp. 31–32] **[TBD-2]**	PARTLY: same board as the G474RE
Independent watchdog	MET [5, p. 1557]	MET [5, p. 1557]	MET [5, p. 1557]
RTC with an independent clock source	PARTLY: internal oscillator only, 29.5 to 34 kHz [4, pp. 51, 105]	MET: 32.768 kHz crystal fitted [16, Sec. 6.5.2, p. 22]	MET: same board as the G474RE
Owned	YES	NO	NO

5.0 Onboard Computer

The only candidate is the STM32MP257F-DK, which the author owns. Its 40-pin header carries one UART, USART6, on pins 8 and 10 [8, Table 24, p. 35]. It has two USB Type-A host ports [8, p. 2].

Need [2, Sec. 8.0]	Answer	STM32MP257F-DK
UART for data link A	MET	USART6 on the header [8, Table 24, p. 35]
UART to the factory loader	PARTLY	No second UART documented on the header; a USB-to-RS-422 adapter on a host port meets it **[TBD-3]**
CAN FD controller with transceiver	PARTLY	No CAN signal on the header and no transceiver on the board [8, Table 24, p. 35]; a USB CAN FD adapter meets it, with CAN FD support in the mainline Linux driver [12, ll. 181, 969]
Four GPIOs: SWDIO, SWCLK, NRST, BOOT	MET	Free header pins, for example PZ0, PZ1 and PZ9 on pins 15, 16 and 13 [8, Table 24, p. 35]
Storage for two firmware stores	MET	8 GB eMMC and a microSD slot [8, pp. 2, 18]

6.0 Interface Parts and Bench Tools

No ready-made modules were found for the transceivers below; they are mounted on small adapter boards.

Need	Candidate	Count	Facts
RS-422 line drivers, data link A and factory loader	TI THVD1451	4	3 to 5.5 V supply; full duplex; up to 50 Mbps [9, p. 1; Table 5, p. 4]
CAN FD transceiver	TI TCAN332G	2	Single 3.3 V supply; CAN FD up to 5 Mbps [10, p. 1]
Debug port shared by the probe and the onboard computer [3, Sec. 5.0]	TI TMUX1574	1	Four 2:1 switches, so SWDIO, SWCLK, NRST and BOOT change over together; 1.5 to 5.5 V supply [11, p. 1]
Controller power cut at a chosen moment [3, Sec. 5.0]	USB hub with per-port power switching	1	Switched from a script by `uhubctl` [13, Sec. What is USB per-port power switching?]; the hub model **[TBD-4]**
Events time-stamped from one clock [3, Sec. 5.0]	Logic analyzer	1	Model **[TBD-5]**
Data link cut and bytes changed on command [3, Sec. 5.0]	Board in the middle of the link	1	Board **[TBD-6]**

7.0 Selection

7.1 First Bench

The first bench uses only the two owned boards (Figure 1). The NUCLEO-G431KB's USB plugs into a host port of the STM32MP257F-DK. Through the on-board ST-LINK, one cable gives the onboard computer the controller's USART2 as data link A [6, p. 16] and the controller's SWD port for the restore. Power is cut by hand, and events are time-stamped by the onboard computer's clock. It costs nothing.



Figure 1: The first bench: owned hardware only.

It shows the update, the trial, the return to the previous copy, the refusal of damaged and forged images, and the restore of an erased controller. It does not show data link B, the RS-422 and CAN electrical links, the restore through the factory loader, SWD driven from the onboard computer's own pins, or power cuts made by a script.

7.2 Full Bench

The full bench adds the parts of Section 6.0: four THVD1451, two TCAN332G, one TMUX1574, their adapter boards, a switching USB hub and a logic analyzer. On the onboard computer, a USB-to-RS-422 adapter carries the factory loader's UART and a USB CAN FD adapter carries data link B, unless the STM32MP257 datasheet shows both on header pins [**TBR-1**].

The controller stays the NUCLEO-G431KB, which the author owns; the NUCLEO-G474RE is not available. On the full bench, its BOOT0 pin and its SWD pins are wired out from the board, since none of them reaches a header [6, pp. 16, 22–23]. Its single flash bank stalls the running code for up to 24.47 ms for each page erased while an update is written [5, p. 195], a limit on delivering firmware while the controller keeps working.

A Acronyms

CAN FD Controller Area Network with Flexible Data rate
GPIO General-purpose input/output
NRST Reset pin
RTC Real-time clock
SWD Serial Wire Debug
TBD To Be Determined
TBR To Be Resolved
UART Universal asynchronous receiver-transmitter
USB Universal Serial Bus

B Open Items

The values below are not yet known.

TBD-1 Project name and document reference.
TBD-2 The board change that connects BOOT0 to header CN7 on the Nucleo-64 boards; settled by the board schematic.
TBD-3 Whether the STM32MP257's pins can put a second UART and CAN FD on the header; settled by its datasheet.
TBD-4 The USB hub model.
TBD-5 The logic analyzer model.
TBD-6 The board placed in the middle of a data link.

Accepted by the author on 12 October 2026: the first bench (Section 7.1); the NUCLEO-G431KB as the controller (Section 7.2). The proposal below awaits the author's acceptance.

TBR-1 The parts of the full bench (Section 7.2).